ONBRD-06: Organizing Your Environment

> **Series:** ONBRD — Dynatrace Onboarding | **Notebook:** 6 of 10 | **Created:** December 2025 | **Last Updated:** 10/14/2026

Tags, Segments, and Naming Conventions

As your Dynatrace environment grows, organization becomes critical. This notebook covers how to structure your environment with tags, segments, and naming conventions for maintainability and access control.

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Prerequisites

- Admin or Configurator access to Dynatrace
- Entities discovered (hosts, services)
- Understanding of your organizational structure

OneAgent Attribute Enrichment (OneAgent 1.333+)

> **Requires:** OneAgent version **1.333** or later

OneAgent can enrich **all telemetry signals** (metrics, spans, logs, events, entities) with custom metadata at the source — before data reaches the Dynatrace platform. This is more efficient than server-side tagging (auto-tags) because enrichment happens on the host and propagates to all Smartscape nodes.

Primary Fields (standardized from Semantic Dictionary):
- `dt.security_context` — data governance and access control
- `dt.cost.costcenter` — cost allocation
- `dt.cost.product` — product attribution

Primary Tags (custom key-value pairs):
- `primary_tags.environment` — environment identification (production, staging, etc.)
- `primary_tags.team` — team ownership
- `primary_tags.business_unit` — organizational unit

Configuration:

```
# Set during OneAgent installation
Dynatrace-OneAgent-Linux.sh --set-host-
tag="primary_tags.environment=production" --set-host-
tag="dt.security_context=confidential"

# Set on existing agents via oneagentctl
oneagentctl --set-host-tag="primary_tags.environment=production"
oneagentctl --set-host-tag="dt.cost.costcenter=12345"

# Per-process via environment variable (overrides host-level)
DT_TAGS="primary_tags.team=platform primary_tags.environment=production"
```

Benefits over auto-tagging:

Aspect	Auto-Tags (Server-Side)	Attribute Enrichment (Agent-Side)
When applied	After data arrives at platform	At the source, before transmission
Scope	Entity tags only	All signals: metrics, spans, logs, events, entities
Grail integration	Limited	Full — feeds OpenPipeline routing, bucket assignment, permissions

| Cost allocation | Not supported | `dt.cost.costcenter`, `dt.cost.product` fields |
| Security context | Not supported | `dt.security_context` for data governance |

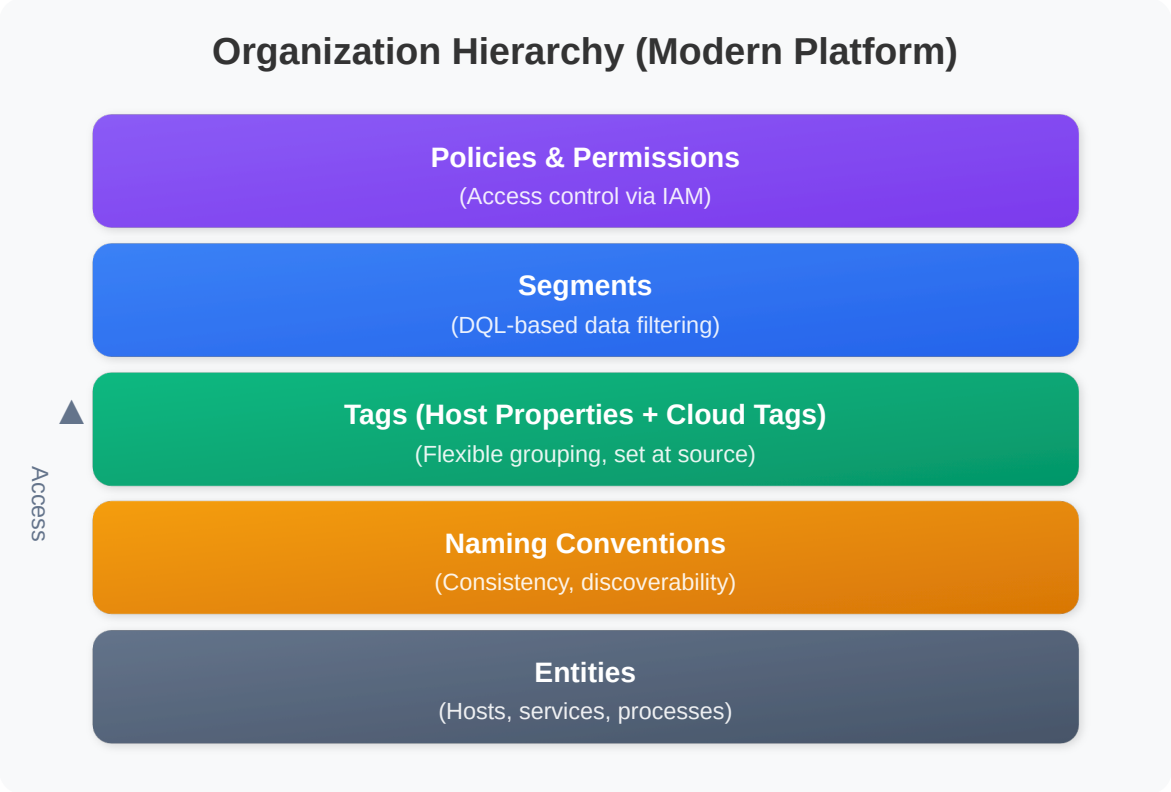
> **See:** Primary Grail fields and tags enrichment through OneAgent
>
> **Version Support Policy:** OneAgent and ActiveGate versions are supported for **9 months (Standard)** or **12 months (Enterprise)**. Third-party technologies are supported for 6 months beyond vendor EOL. See Support Policy.

1. Why Organization Matters

Without organization, Dynatrace environments become difficult to manage:

Problem	Impact
No structure	Can't find entities quickly
No ownership	Don't know who to contact
No access control	Everyone sees everything
No filtering	Dashboards show irrelevant data
No grouping	Can't compare similar systems

Modern Organization Building Blocks



> **Note:** The modern Dynatrace platform uses **Segments** for data filtering and **Policies** for access control. This replaces the legacy Management Zones approach.

2. Modern Organization Building Blocks

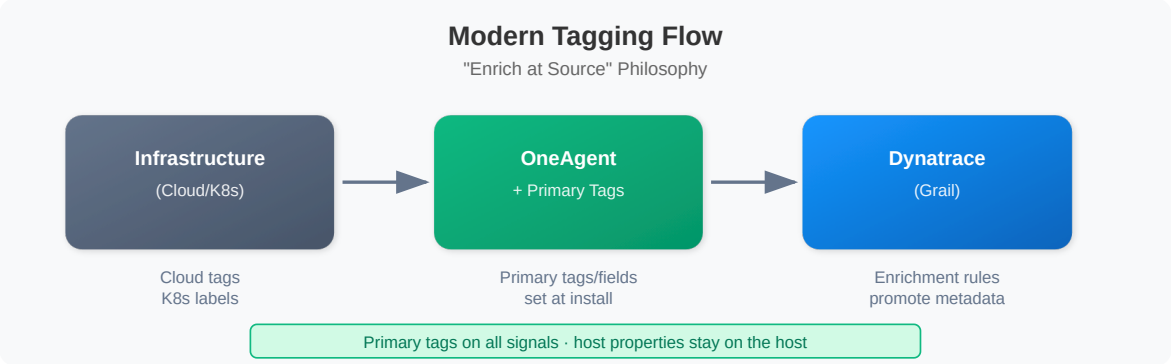
The modern Dynatrace platform (Gen3/Grail) uses a "tag at source" approach rather than rule-based auto-tagging:

Tag Sources

Source	How It Works	Best For
Primary tags / primary fields	Set at OneAgent install or with `oneagentctl --set-host-tag` (OneAgent 1.333+)	Environment, team, security context, cost allocation — on all signals

Source	How It Works	Best For
Host Properties	Set with `--set-host-property`; stay on the host node unless an ingest enrichment rule promotes them	Host-level metadata
Cloud Provider Tags	Promoted onto signals by ingest enrichment rules (OneAgent / ActiveGate 1.343+; cloud rule types rolling out)	Cloud resource organization
Kubernetes Labels	Promoted onto signals by ingest enrichment rules (Dynatrace Operator 1.10+ with `metadataEnrichment`)	Container workload organization
OpenTelemetry Attributes	Set in instrumentation	Custom service attributes

The "Enrich at Source" Philosophy



Why "Tag at Source"?

Benefit	Description
Consistent	Primary tags and fields land on metrics, logs, spans, events, and entities
Scalable	No processing overhead to apply rules
Traceable	Tags come from the source of truth
Real-time	No delay waiting for rule evaluation

3. Tagging with Host Properties and Cloud Tags

Host Properties (OneAgent)

Set custom properties during OneAgent installation or via configuration:

During Installation:

```
# Linux
sudo /bin/sh Dynatrace-OneAgent.sh \
  --set-host-property=env=production \
  --set-host-property=team=platform \
  --set-host-property=tier=backend

# Windows
.\Dynatrace-OneAgent.exe --set-host-property=env=production --set-host-
property=team=checkout
```

After Installation:

Change properties with `oneagentctl` (OneAgent 1.189+), or centrally with Remote configuration management:

```
./oneagentctl --set-host-property=env=production --set-host-
property=team=platform
```

The `hostcustomproperties.conf` file applies only to OneAgent 1.187 and earlier.

> **Where host properties land:** a host property is metadata on the HOST node — on Smartscape it appears in the `host.custom.metadata` record — not a field on logs, spans, or metrics. To filter signals by a value, set it as a primary tag instead (`--set-host-tag="primary_tags.<key>=<value>"` , see Prerequisites above) or promote the property with a Host/Process property ingest enrichment rule (DT docs).

Recommended Property Categories

Category	Example Properties	Purpose
Environment	`env=prod` , `env=staging` , `env=dev`	Distinguish environments

Category	Example Properties	Purpose
Owner	`team=platform`, `team=checkout`	Identify responsible team
Application	`app=ecommerce`, `app=mobile-api`	Group by application
Cost Center	`cost-center=marketing`	Financial allocation
Tier	`tier=frontend`, `tier=backend`	Architecture layer

Cloud Provider Tags

Cloud tags reach your telemetry through **ingest enrichment rules**. Cloud tag rules need a new cloud connection and Dynatrace 1.348+; the OneAgent route below needs OneAgent 1.343+. Cloud rule types are rolling out and may not yet be visible in your tenant — until a rule exists, the raw provider tags are visible on the HOST node (for example a `tags:azure` record on an Azure VM) but are not fields on logs, spans, or metrics.

Cloud	Tag Source	Attribute Written to Telemetry by a Rule
AWS	AWS resource tags	`aws.tags.<key>`
Azure	Azure resource tags	`azure.tags.<key>`
GCP	Google Cloud labels / tags	`gcp.labels.<key>` / `gcp.tags.<key>`

Which data a cloud tag rule reaches. A cloud tag rule such as an AWS tag rule tags only the data that arrives through the cloud connection. That means cloud metrics, cloud logs and the cloud Smartscape nodes. In Dynatrace's words, *"OneAgent on a cloud VM doesn't evaluate them."* A VM that runs OneAgent therefore needs a second rule for its OneAgent data:

Data from an EC2 instance	Rule	Notes
Cloud metrics, cloud logs, the `AWS_EC2_INSTANCE` node	AWS tag rule at the AWS account scope	Writes `aws.tags.<key>`, or a security context or cost field. Primary tag isn't a target yet, so promote `aws.tags.<key>` with an OpenPipeline primary Grail tag rule (FAQ-26 §5.4)

Data from an EC2 instance	Rule	Notes
Logs, spans and metrics from OneAgent on the instance	Host/Process property rule at the host-group scope, with source `aws.tags.<key>`	OneAgent reads instance tags from instance metadata. *"Access to instance tags in instance metadata is off by default in AWS, so enable it on the instance first."*

AWS Tag Example:
An EC2 instance is tagged `team=checkout`. An AWS tag rule for `team` adds `aws.tags.team` to the instance's cloud metrics and logs. A Host/Process property rule with source `aws.tags.team` and target Primary tag `team` writes `primary_tags.team` on everything OneAgent sends from that instance. Either rule can instead write the value to a primary field such as Security context or Cost center. Worked end to end in Apply primary Grail tags across an example AWS estate (DT docs).

Kubernetes Labels

Namespace and workload names are set on Kubernetes telemetry. Labels reach signals only through ingest enrichment rules (Dynatrace Operator 1.10+ with `metadataEnrichment` enabled in the DynaKube). Namespace-label rules are available now; workload and pod rule types are rolling out.

K8s Metadata	DQL Field
Namespace	`k8s.namespace.name`
Deployment	`k8s.deployment.name`
Namespace labels	`k8s.namespace.label.<key>` (enrichment rule)
Workload labels	`k8s.workload.label.<key>` (enrichment rule)
Pod labels	`k8s.pod.label.<key>` (enrichment rule)

4. Segments for Data Filtering

Segments provide reusable filters that create focused views of your data.

Location: the segment selector in any app → **Manage segments**

What are Segments?

Segments are reusable filters built from per-data-type **includes** that:
- Filter data in Notebooks, Dashboards, and Apps
- Can be applied as default context
- Are shareable across the organization

Creating a Segment

1. Open the segment selector in any app → **Manage segments** → **Segment**
2. Name your segment (e.g., "Production Environment") and add a description
3. Add an **include**: choose a data type — or **All data types** — and a filter condition, for example:
   `text primary_tags.environment = production`
4. Select **Run query** to preview matching data
5. Save

> **Includes decide what a segment returns.** Querying a data type that no include references returns **empty results** while the segment is applied — use *All data types* when one condition should apply everywhere.

Segment Use Cases

Use Case	Include Condition	Purpose
Environment	`primary_tags.environment = production`	Focus on production
Team	`primary_tags.team = checkout`	Team-specific view
Kubernetes stack	`k8s.cluster.name = gke-klu`	Cluster-group focus
Region	`aws.region = us-east-1`	Geographic filtering

> **Conditions need fields that exist on the data.** `primary_tags.*` matches only data enriched at source (see Prerequisites above). Host properties set with `--set-host-property` are not fields on signals, so they cannot drive a segment until an enrichment rule promotes them.

Segments vs Legacy Management Zones

Feature	Segments	Management Zones (Legacy)
Filter basis	Per-data-type include conditions	Rule-based matching
Data types	All Grail data	Entities only
Flexibility	Highly flexible	Limited rule types
Access control	Use Policies + `dt.security_context` instead	Built-in
Modern platform	✅ Recommended	⚠️ Dynatrace Classic concept — e.g. API 1.337 deprecated the `managementZone` property on calculated service metrics

> **Where to go deeper:** the **ORGNZ series** (11 notebooks) covers segments, buckets, and `dt.security_context` design in depth. The **IAM series** (especially IAM-04, IAM-05, IAM-11 WORKSHOP) covers how policies use `dt.security_context` to scope access. **MZ2POL** (11 notebooks) covers Management Zone → Policy migration if you have legacy MZs to retire.

Segment Best Practices

Practice	Why
Use primary tags / `dt.security_context`	Consistent filtering; aligns with IAM scoping
Name clearly	`Prod-Checkout-Team` not `Segment1`
Document purpose	Add description
Test filters	Verify expected data

5. Naming Conventions

Consistent naming makes entities discoverable and filtering effective.

Host Naming

Set meaningful host names that encode key information:

Pattern	Example	Components
`{env}-{tier}-{seq}`	`prod-web-01`	Environment, tier, sequence
`{region}-{app}-{role}`	`us-east-ecom-api`	Region, app, role
`{team}-{service}-{id}`	`checkout-cart-a1b2`	Team, service, unique ID

Host Naming via OneAgent

You can set a custom display name during installation:

```
sudo /bin/sh Dynatrace-OneAgent.sh --set-host-name="prod-web-01"
```

Naming Principles

Principle	Good	Bad
Descriptive	`payment-service`	`svc-001`
Consistent	`prod-web-01`, `prod-web-02`	`prod-web-01`, `Web Server 2`
Parseable	`us-east-prod-checkout`	`USEastProdCheckout`
Unique	Include environment/region	Generic names

Property Naming Standards

For host properties, use consistent naming:

Standard	Example	Why
Lowercase	`env=prod` not `ENV=PROD`	Consistency
Hyphen separated	`cost-center=eng`	Readability
Short keys	`env` not `environment`	Query simplicity
Consistent values	Always `prod` not sometimes `production`	Filtering works

6. Querying by Tags and Properties

Use primary fields, primary tags, and host properties in DQL queries to filter and group data. The first two queries read tag data directly; the rest filter on names and built-in dimensions.

```
// Log volume by security context (a primary field set at source)
fetch logs, from: now() - 1h
| filter isNotNull(dt.security_context)
| summarize log_count = count(), by: {dt.security_context}
| sort log_count desc
| limit 20
// Same pattern for a primary tag set at install, e.g.:
//   | filter primary_tags.environment == "production"
```

```
// Host group and host properties on HOST nodes
// (host properties live in the host.custom.metadata record, not on signals)
smartscapeNodes "HOST", from:-7d
| fields name, dt.host_group.id, host.custom.metadata
| sort name
| limit 20
// Filter on one property, e.g.:  | filter host.custom.metadata[env] ==
"production"
```

```
// Find hosts by name pattern
fetch dt.entity.host
| filter contains(entity.name, "prod")
| fields entity.name
| limit 20

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | filter contains(name, "prod")
//   | fields name
//   | limit 20
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: entity.name -&gt; name.
```

```
// Count hosts by operating system type
fetch dt.entity.host
| summarize host_count = count(), by: {osType}
| sort host_count desc

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | summarize host_count = count(), by: {os.type}
//   | sort host_count desc
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: osType -&gt; os.type (LINUX -&gt; OS_TYPE_LINUX).
```

```
// Find services by name pattern
fetch dt.entity.service
| filter contains(entity.name, "checkout")
| fields entity.name, serviceType
| limit 20

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "SERVICE"
//   | filter contains(name, "checkout")
//   | fields name, dt.service.sdv1_type
//   | limit 20
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: serviceType -&gt; dt.service.sdv1_type (SDv1 services only -
null when
// dt.service_detection.version == 2; group by dt.service_detection.version to
see
// which model applies); entity.name -&gt; name.
```

```
// Log volume by Kubernetes namespace
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize log_count = count(), by: {k8s.namespace.name}
| sort log_count desc
| limit 20
```

```
// Query spans by service name pattern
// dt.service.name is set on every span, whatever the ingest source
fetch spans, from: now() - 1h
| filter span.kind == "server"
| filter contains(dt.service.name, "payment")
| summarize request_count = count(), by: {dt.service.name}
| sort request_count desc
| limit 20
```

```
// Find hosts by name pattern for environment identification
fetch dt.entity.host
| filter not(contains(entity.name, "prod"))
      and not(contains(entity.name, "staging"))
      and not(contains(entity.name, "dev"))
| fields entity.name
| limit 20

// Smartscape equivalent (dt.entity.* is deprecated but still functional):
//   smartscapeNodes "HOST"
//   | filter not(contains(name, "prod"))
//         and not(contains(name, "staging"))
//         and not(contains(name, "dev"))
//   | fields name
//   | limit 20
// Caveat: Smartscape can report fewer entities than the classic entity store,
and
// both return only entities seen in the query timeframe (default 2 h) - for a
// discovery inventory add a timeframe such as from:-7d to either query.
// Field maps: entity.name -&gt; name.
```

7. Next Steps

With organization in place:

1. **ONBRD-07: Understanding Your Data** — Explore what Dynatrace discovered
2. Define primary tags (and any host properties) for your environment
3. Create segments for team-specific views
4. Document your naming conventions
5. Decide your `dt.security_context` value space (and bind IAM policies to it)

Where to Go Deeper

- **ORGNZ series** (11 notebooks) — Bucket strategy, segments, security context, the full data-organization model
- **FAQ-01** — Host group naming strategy
- **FAQ-02** — Tagging sources, standards, and strategy (primary tags vs cloud tags vs auto-tags)
- **IAM series** (especially IAM-04, IAM-05, IAM-11 WORKSHOP) — Policy design that consumes `dt.security_context`
- **MZ2POL series** (11 notebooks) — Management Zone → Policy migration for legacy MZ retirement

Organization Checklist

- [ ] Host property strategy documented
- [ ] Primary tags / primary fields set at OneAgent install (`--set-host-tag="primary_tags.…"`, OneAgent 1.333+, see ONBRD-05)
- [ ] Cloud tags verified (if using cloud providers; see FAQ-02 for cloud-tag normalization)
- [ ] `dt.security_context` value space decided
- [ ] Segments created for common filters
- [ ] Naming conventions established (FAQ-01 for host groups)
- [ ] Access control configured via Policies (see ONBRD-02 + IAM series)

Summary

In this notebook, you learned:

- Why organization matters for scalability
- The modern "tag at source" approach (primary fields + primary tags via OneAgent)
- Where host properties, cloud tags, and K8s labels land — and how ingest enrichment rules promote them onto signals
- How to create and use Segments built from per-data-type includes
- Naming convention best practices

- How `dt.security_context` standardizes the boundary field for Gen3 IAM
- How to query by properties and attributes

References

- Host Properties
- Segments
- Cloud platform monitoring (DT docs)
- Kubernetes Labels
- DQL Reference
- OneAgent Attribute Enrichment
- Configure ingest enrichment rules (DT docs) — *"Cloud rule types are rolling out and may not yet be visible in your environment."*
- Define tags and metadata for hosts (DT docs)
- OneAgent configuration via command-line interface (DT docs) — *"For versions earlier than 1.189, use a host metadata configuration file."*
- Include data in segments (DT docs) — *"Querying for data not explicitly referenced by any include of the selected segment, will lead to empty results."*
- Segments for Kubernetes clusters (DT docs)
- Dynatrace API 1.337 (DT docs)

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.